

the time and cost associated with physical validation. This shift towards virtualisation is not just a response to the demands of rapid product development; it is a necessary evolution in an industry that is constantly pushing the boundaries of technology and innovation.

Better vehicle design with smart models

Model-based systems engineering (MBSE) is revolutionising the way we develop products. At the heart of MBSE is the use of models as the central element of the development process. These models are virtual replicas of components, subsystems, or even the entire vehicle. They provide a comprehensive understanding of how the product will behave in real-world scenarios.

The beauty of MBSE lies in the evolution of these models. They are not static; they grow and adapt as more information becomes available throughout the product development cycle. For instance, at the beginning of the development process, I might only know the basic specifications of a motor. But as the design progresses, my understanding of the motor's characteristics improves, and the model evolves accordingly.

To illustrate this, let us consider the standard V-cycle of product development, a framework widely recognised across the industry. The process begins with defining the requirements based on stakeholder needs, such as the desired acceleration or range of a vehicle. Using the MBSE approach, we then systematically develop the vehicle architecture, focusing on the systems of interest, which are the key areas for innovation in each design cycle.

Concurrent engineering comes into play as we delve into the specifics. For example, while the battery development team works on the cell and pack design, a parallel team focuses on the software aspects of the

battery management system. This approach ensures that both hardware and software components are developed in tandem, optimising the overall system performance.

As we move further into the detailed design stage, the models become more refined, incorporating intricate details such as cooling circuits and cell placement. These designs are then integrated back into the system model for validation at the subsystem level, using model-in-loop and hardware-in-loop environments.

The final stage involves vehicle-level performance evaluation where key performance parameters of the vehicle are evaluated and compared against requirements. The virtualisation process, however, does not stop there. With the advent of digital twins, we can now monitor the vehicle's health in real-world conditions, enabling predictive maintenance and diagnostics.

The systematic nature of MBSE is particularly valuable in managing the complex, interdependent, and often conflicting requirements of vehicle development. For example, balancing performance with cost constraints can be challenging, but MBSE provides a structured approach to navigate these trade-offs.

Not every organisation has fully embraced MBSE yet, but for those who do find it immensely beneficial, it allows for continuous validation at every stage of development, preventing costly mistakes and rework. Various analytical tools, from simple Excel sheets to advanced simulation models, are used to support this process, ensuring that the vehicle's performance aligns with the defined requirements.

Testing products with simulations

Virtual validation is a transformative approach in product development that utilises simulations to predict a product's performance before physi-



**Established
Since 1977**

**Import & Stockist
Pan India Agent**



RANGE OF PRODUCTS

LAMINATES

1. FR1 (Paper based Copper Clad Laminates)
2. FR4 (Glass Based Copper Clad Laminates)
3. CEM1 (Composite based Copper Clad Laminates)
4. MCCL (Aluminium based Copper Clad Laminates)

BACK UP BOARD

1. Melamine Based Backup Board
2. HDF Back up Board
3. Aluminium Entry Sheet

OTHERS

1. PCB Inks (UV, PISM, THERMAL, LED)
2. Copper Measuring Instrument
3. Karlson Printing Pin
4. V Grooving Blade
5. Drill Bits

Contact Details

Ashish Shah : +91 - 99207 84787
Mayank shah : +91 - 98200 28773

Email Id

nmbinita@gmail.com

Address

304, Phoenix building, 457 S.V.P. Road,
Mumbai 400004